

DATABASE FUNDAMENTALS

ERD Assignment

Entity Relationship Diagram Design

Topics Covered in This Unit

- What is a database? • Database vs. files • Schema
- Tables, rows, and columns • Entities and attributes • Primary keys
- Relationships • Entity Relationship Diagrams (ERD)
- Relationship cardinality: One-to-One (1:1), One-to-Many (1:M), Many-to-Many (M:N)

Instructions

This assignment contains **two independent scenarios**. For each scenario, read the description carefully and design a complete Entity Relationship Diagram (ERD) that models the system described. You must work out the entities, attributes, primary keys, relationships, and cardinalities yourself — no solution structure is provided below. Submit your work as a single PDF following the file naming convention at the end of this document.

Cardinality notation used throughout this assignment:

1:1	One-to-One
1:M	One-to-Many
M:N	Many-to-Many

Hospital Management System

Scenario

A hospital wants to develop a database system to manage its departments, doctors, patients, appointments, medical records, and treatments.

The hospital has several departments. Each department has a unique ID, name, and location. A department can have many doctors, but each doctor works in only one department.

Each doctor has a unique ID, name, specialization, phone number, and email.

The hospital has many patients. Each patient has a unique ID, name, date of birth, gender, phone number, and address.

A patient can make appointments with doctors. A doctor can have appointments with many patients. For every appointment, the hospital needs to store the appointment date, appointment time, status, and reason for the visit.

A patient can have multiple medical records. Each medical record belongs to one patient and contains the record date, diagnosis, notes, and treatment plan.

A medical record can contain multiple treatments. The same treatment can be used in many different medical records. Each treatment has a unique ID, name, description, and cost.

Required

Create a complete ERD for the hospital management system. Your ERD must include:

- 1 All required entities.
- 2 All relevant attributes.
- 3 A primary key for each entity.
- 4 All relationships.
- 5 The cardinality of every relationship.

For every relationship, specify whether it is One-to-One (1:1), One-to-Many (1:M), or Many-to-Many (M:N).

Questions

Answer only the following two questions:

1. How does your ERD represent the relationship between patients and doctors through appointments?
2. How does your ERD represent the relationship between medical records and treatments?

Hotel Reservation & Management System

Scenario

A hotel wants to develop a database system to manage its guests, rooms, room types, reservations, services, and payments.

The hotel has many rooms. Each room has a unique room number, floor number, and status. Every room belongs to one room type.

A room type describes the characteristics of the room, such as its name, description, maximum number of guests, and price per night. One room type can be assigned to many rooms.

The hotel has guests. Each guest has a unique ID, name, email, phone number, nationality, and date of birth.

A guest can make many reservations. Each reservation belongs to one guest and contains a reservation ID, booking date, check-in date, check-out date, number of guests, and reservation status.

A reservation can include one or more rooms, and a room can appear in many reservations over time. For each room included in a reservation, the hotel needs to store the price per night agreed upon for that reservation.

During their stay, guests can request additional hotel services such as room service, laundry, transportation, or cleaning.

A reservation can have multiple service requests, and the same service can be requested in many different reservations. For each service request, the hotel needs to know the quantity requested and the date and time of the request.

Each service has a unique ID, name, description, and price.

A reservation can have one or more payments. Each payment has a unique payment ID, payment date, amount, payment method, and payment status.

Required

Create a complete ERD for the hotel reservation and management system. Your ERD must include:

- 1 All required entities.
- 2 All relevant attributes.
- 3 A primary key for each entity.
- 4 All relationships.
- 5 The cardinality of every relationship.

For every relationship, specify whether it is One-to-One (1:1), One-to-Many (1:M), or Many-to-Many (M:N).

Questions

Answer only the following two questions:

1. How does your ERD represent the relationship between reservations and rooms, considering that a reservation can contain multiple rooms and a room can appear in multiple reservations over time?
 2. How does your ERD represent the relationship between reservations and services, considering that a reservation can request multiple services and the same service can be requested in many reservations?
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Submission

Submit one PDF containing both assignments. The PDF must contain:

Assignment 1	Assignment 2
Hospital Management System ERD	Hotel Reservation & Management System ERD
Cardinality of all relationships	Cardinality of all relationships
Answers to the two questions	Answers to the two questions

File Naming Format

StudentName_GroupDay_ERD_Assignment9.pdf